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Prof. Guangtao Zhai
Editor-in-Chief
Displays
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Dear Professor Zhai,

We transfer the manuscript “**Uncollapsed and Metrically-Scaled 3D Reconstruction from Monocular Endoscopic Video via Pseudo-3D Heightfields**” to *Displays* as a Research Article. The manuscript asks a display-first question: can monocular endoscopic video be turned into a 3D surface that is simultaneously *displayable* (un-collapsed, renderable) and *measurable* (metrically scaled)? We show that the answer is a pseudo-3D heightfield used in two roles—as a geometric matching substrate and as an operator corrector.

Why Displays. The useful output is a renderable mesh for intra-operative overlay and measurement, not a disparity map alone. A collapsed cone or planar slab cannot be visualized as a surgical surface and cannot be used to measure physical lengths. This addresses a fundamental display problem: visual signal processing, an un-collapsed 3D surface, and explicit metric scale recovery.

The story in one paragraph. Feed-forward 3D foundation models (VGGT, DUS3R, MAS3R, MUS3R) produce dense but geometrically collapsed, scale-less clouds on endoscopic video, and classical SfM drifts or mis-decomposes under small planar baselines. We reconstruct a bilateral gradient-anchored pseudo-3D heightfield from each frame and exploit it twice. As a *geometric matching substrate*, it registers frames (FPFH/RANSAC/ICP), builds dense epipolar correspondences, and feeds a multi-stride bundle adjustment that suppresses dead-reckoning drift. As an *operator corrector*, its physical normal field vetoes the wrong homography branch, and a calibration gauge injects true metric scale. The same mesh therefore drives matching, pose, branch correction, and scale—so the surface stays three-dimensional and metric where the baselines collapse.

Headline results.

- **Un-collapsed Normal 3D geometry:** On the endoscopic chessboard benchmark the heightfield is the only *monocular* reconstruction that stays Normal 3D ($\lambda_3/\lambda_1 = 0.42$); VGGT, MAS3R, DUS3R, MUS3R, COLMAP and ORB-SfM all collapse to Flat or Cone (≤ 0.22), and only the stereo SGM reference also reaches Normal 3D (0.31).
- **Near-ideal metric scale:** The gauge recovers $1.11\times$ scale against the reference, while every baseline drifts by $3.7\times$ – $647\times$.

- **Leading depth on public data:** On EndoNeRF the calibrated stereo variant attains the best AbsRel (0.0285), ahead of FoundationStereo (0.0321) and Stereo SGM (0.0323).
- **Drift suppression and gauge ceiling:** On a 17-sequence benchmark the pattern-free multi-view BA outperforms ORB-SfM (35.52 vs. 43.72 mm mean ATE) and beats VGGT-1B in a smoke-filled cavity (24.86 vs. 37.25 mm); with the gauge visible, the gauge-referenced trajectory reaches the 4.49 mm pattern-localisation floor.

Declarations. This manuscript is not under consideration elsewhere. All authors have approved the submission. The study uses public datasets and in-house bench-top phantom recordings; it does not involve new human-subject or animal experiments. The authors have no competing interests. Code, evaluation scripts, and result files are at <https://github.com/liranyang123456-commits/re3d-heightfield>. Full in-house sequences remain available from the corresponding author.

We thank you for considering this work.

Sincerely,

Ranyang Li, et al.